

Biomonitoring of Carcinogenic Heterocyclic Aromatic Amines in Hair: A Validation Study



A facile method was established to measure heterocyclic aromatic amines (HAAs) accumulated in human hair and rodent fur. The samples were digested by base hydrolysis, and the liberated HAAs were isolated by tandem solvent/solid-phase extraction. Quantification was done by liquid chromatography/tandem mass spectrometry, using a triple stage quadrupole mass spectrometer in the selected reaction monitoring mode. In a pilot study of 12 human volunteers, 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) was detected in hair of six meat-eaters at levels ranging from 290 to 890 pg/g hair. 2-Amino-3,8-dimethylimidazo[4,5-f]quinoxaline (MeIQx) and 2-amino-9H-pyrido[2,3-b]indole (A α C) were below the limit of quantification (LOQ) (50 pg/g hair) in hair from meat-eaters and six vegetarians. PhIP was detected in the hair from one vegetarian, and at level just above the LOQ (65 pg/g hair), indicating PhIP exposure occurs primarily through meat consumption. The levels of PhIP in hair samples from two meat-eaters varied by less than 24% over a 6-month interval, signifying that the exposure to PhIP and its accumulation in hair are relatively constant over time. In a controlled feeding study, female C57BL/6 mice were given these HAAs in their drinking water for 1 month, at six daily dose concentrations ranging from 0, 0.080 to 800 μ g/kg body weight. PhIP was detected in fur of mice at all doses, whereas A α C and MeIQx were detected in fur at dosages ≥ 0.8 μ g A α C/kg body weight and ≥ 8 μ g MeIQx/kg body weight. There was a strong positive relationship between dosage and each of the HAAs accumulated in fur and their DNA adducts formed in liver and colon (p-values < 0.0001); however, the levels of HAA in fur did not correlate to the levels of DNA adducts after adjustment of dose. Thus, hair appears to be a promising long-lived biomarker with by which we can assess the exposure to PhIP, a potential human carcinogen.